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DEPARTMENT OF INFORMATION TECHNOLOGY

Mini Project ITL601 Business Intelligence LAB

Title of the Project

Sales Data Analysis using Star & Snowflake Schema (Business Intelligence Approach)

Supervisor/Guide

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REV - 2019 'C' Scheme

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PROBLEM DEFINITION

In today's competitive business environment, organizations generate large volumes of sales data. However, raw data alone is not sufficient for decision-making. Businesses require structured data models and analytical techniques to extract meaningful insights.

The objective of this project is to:

- Design **Star and Snowflake schemas** for organizing sales data
- Apply **Business Intelligence techniques** to analyze sales performance
- Generate insights that help in **decision-making and business growth**

DATASET DETAILS

The dataset used in this project is a **Sales Dataset** containing transactional records.

Attributes include:

- Order ID
- Order Date
- Customer Name
- Region
- Product Category
- Product Name
- Sales Amount
- Quantity
- Profit

Data Source:

- Sample Sales Dataset (Manual)

Data Characteristics:

- Structured data format (rows & columns)
- Contains both **categorical and numerical data**
- Suitable for **data warehousing and BI analysis**

```
Dataset Preview:
    Order ID  Order Date Region        Category  Sales  Profit  Quantity
0          1  2024-01-01   East       Furniture    200      20         2
1          2  2024-01-05   West      Technology    500      50         5
2          3  2024-02-10  North  Office Supplies    300      30         3
3          4  2024-02-15  South       Furniture    400      40         4
4          5  2024-03-01   East      Technology    600      60         6
```

METHODOLOGY / APPROACH

Step 1: Data Preprocessing

- Removed missing values
- Converted date fields into proper format
- Cleaned inconsistent data

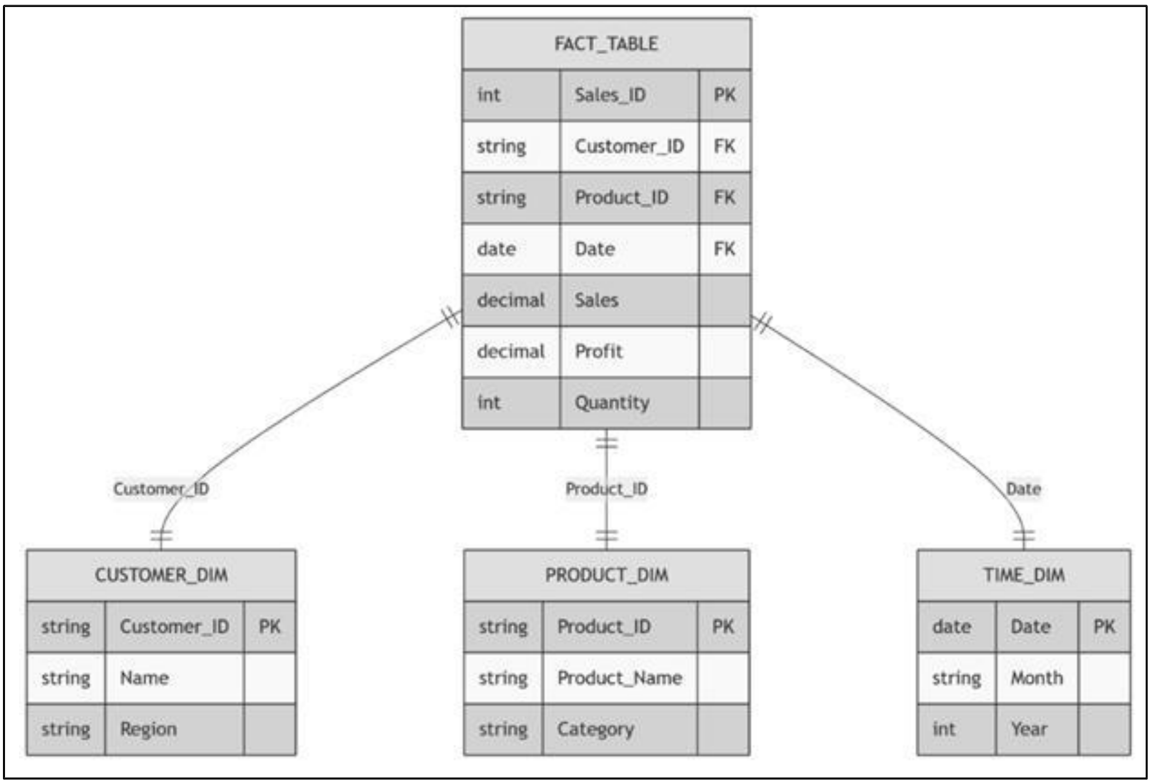
Step 2: Star Schema Design

A **Star Schema** consists of:

- **Fact Table:** Sales Fact (Sales, Profit, Quantity)
- **Dimension Tables:**
    - Customer Dimension
    - Product Dimension
    - Time Dimension
    - Region Dimension

Structure:

- Central Fact Table connected to all dimensions
- Simple and fast for querying



Star Schema Diagram

Step 3: Snowflake Schema Design

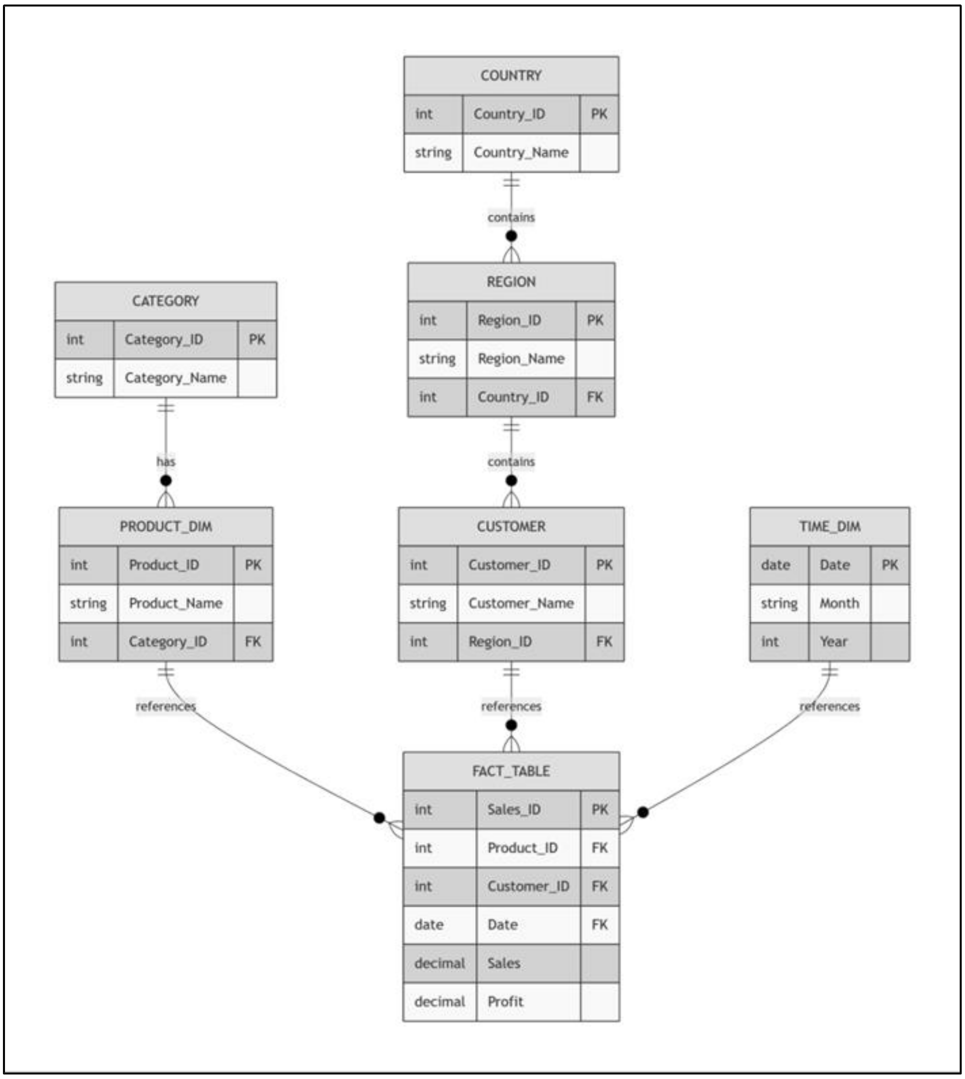
A **Snowflake Schema** is a normalized version of the Star Schema.

Example:

- Product Dimension → split into:
    - Product Table
    - Category Table
- Region → Country → City hierarchy

Advantages:

- Reduces data redundancy
- Improves data integrity



Snowflake Schema Diagram

RESULTS AND VISUALIZATION

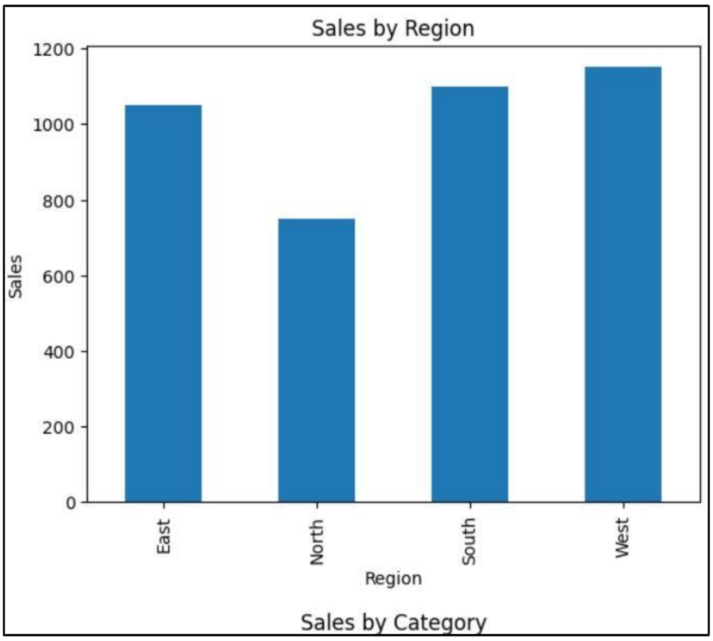
Code :

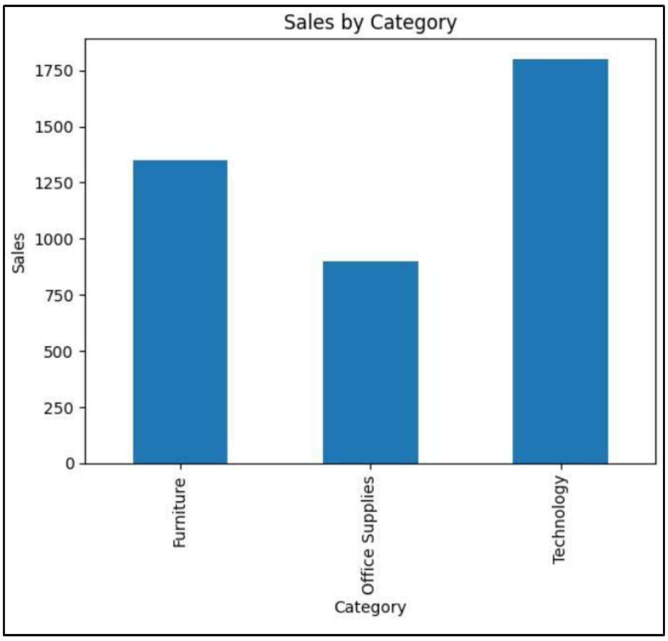
```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# ----------------------------
# STEP 1: CREATE DATASET
# ----------------------------
data = {
    "Order ID": [1,2,3,4,5,6,7,8,9,10],
    "Order Date": ["2024-01-01","2024-01-05","2024-02-10","2024-02-15","2024-03-01",
                   "2024-03-10","2024-04-05","2024-04-12","2024-05-01","2024-05-08"],
    "Region": ["East","West","North","South","East","West","North","South","East","West"],
    "Category": ["Furniture","Technology","Office Supplies","Furniture","Technology",
                 "Office Supplies","Furniture","Technology","Office Supplies","Furniture"],
    "Sales": [200,500,300,400,600,350,450,700,250,300],
    "Profit": [20,50,30,40,60,35,45,70,25,30],
    "Quantity": [2,5,3,4,6,3,4,7,2,3]
}

df = pd.DataFrame(data)

print("Dataset Preview:\n", df.head())

# ----------------------------
# STEP 2: DATA PREPROCESSING
# ----------------------------
df['Order Date'] = pd.to_datetime(df['Order Date'])
df['Month'] = df['Order Date'].dt.month

# ----------------------------
# STEP 3: SALES BY REGION
# ----------------------------
plt.figure()
df.groupby("Region")["Sales"].sum().plot(kind='bar')
plt.title("Sales by Region")
plt.xlabel("Region")
plt.ylabel("Sales")
plt.show()
```

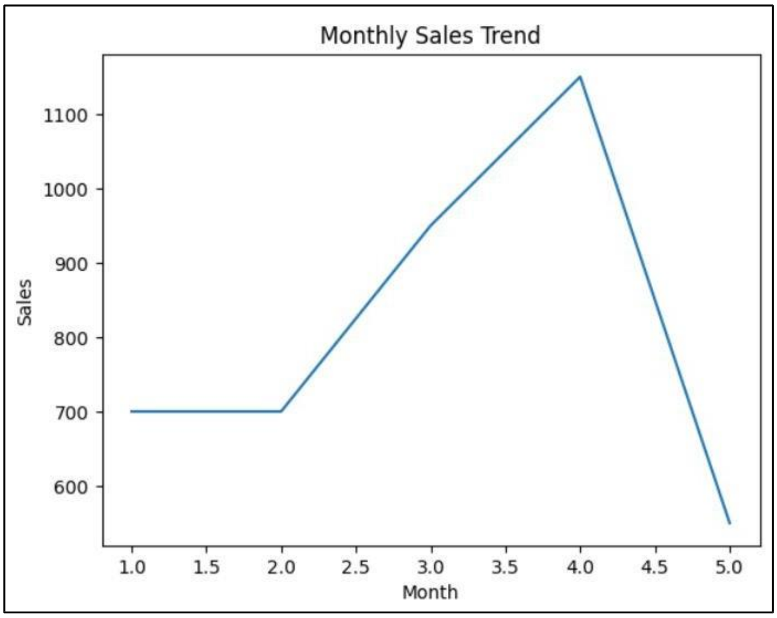
```python
# ----------------------------
# STEP 4: SALES BY CATEGORY
# ----------------------------
plt.figure()
df.groupby("Category")["Sales"].sum().plot(kind='bar')
plt.title("Sales by Category")
plt.xlabel("Category")
plt.ylabel("Sales")
plt.show()

# ----------------------------
# STEP 5: MONTHLY SALES TREND
# ----------------------------
plt.figure()
df.groupby("Month")["Sales"].sum().plot()
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()

# ----------------------------
# STEP 6: PROFIT VS SALES
# ----------------------------
plt.figure()
plt.scatter(df["Sales"], df["Profit"])
plt.title("Profit vs Sales")
plt.xlabel("Sales")
plt.ylabel("Profit")
plt.show()

# ----------------------------
# STEP 7: CORRELATION HEATMAP
# ----------------------------
plt.figure()
sns.heatmap(df.select_dtypes(include='number').corr(), annot=True)
plt.title("Correlation Heatmap")
plt.show()

# ----------------------------
```

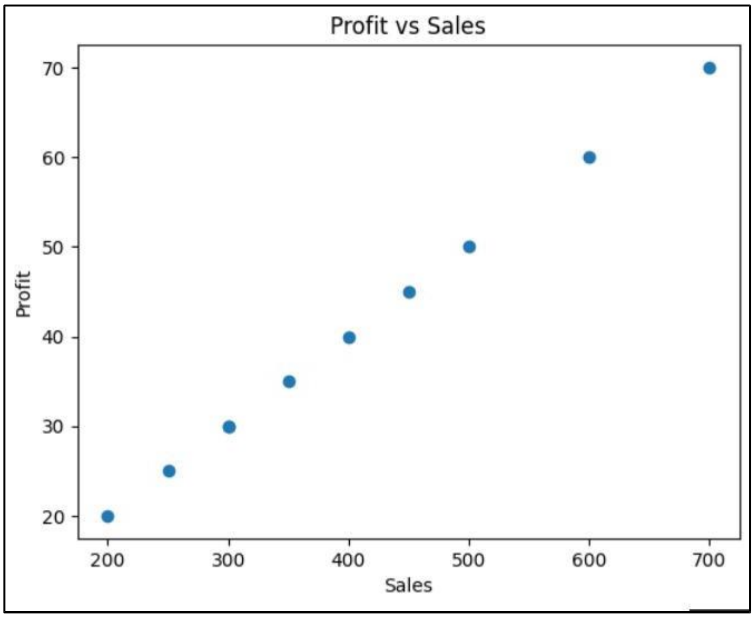
```python
# ----------------------------
# STEP 8: BASIC INSIGHTS
# ----------------------------
print("\nTotal Sales by Region:\n", df.groupby("Region")["Sales"].sum())
print("\nTotal Sales by Category:\n", df.groupby("Category")["Sales"].sum())
print("\nTotal Profit:\n", df["Profit"].sum())
```

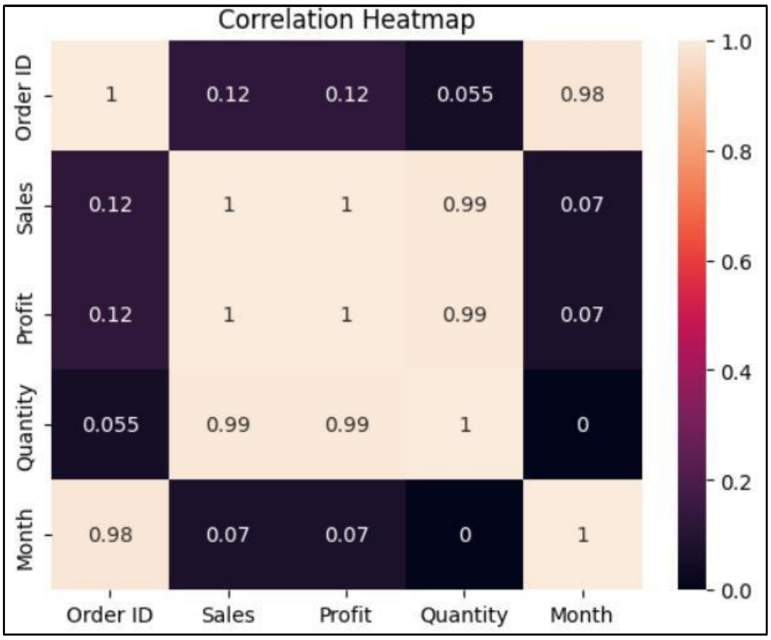
Output :












```
Total Sales by Region:
 Region
East     1050
North     750
South    1100
West     1150
Name: Sales, dtype: int64

Total Sales by Category:
 Category
Furniture          1350
Office Supplies     900
Technology         1800
Name: Sales, dtype: int64

Total Profit:
 405
```

CONCLUSION OR BUSINESS INSIGHTS

Business Insights :

- Some regions contribute more revenue than others
- High sales do not always mean high profit
- Certain product categories dominate the market
- Seasonal trends affect sales performance

Conclusion : This project shows how Business Intelligence techniques and Star & Snowflake schemas help organize and analyze sales data efficiently. The use of visualizations made it easier to identify patterns, trends, and relationships, supporting faster and better decision-making.

The analysis revealed that some regions generate higher sales, while high sales do not always mean high profit. Seasonal trends were observed, helping in planning inventory and marketing. Correlation insights also showed how sales, profit, and quantity are related, enabling businesses to improve performance and profitability.